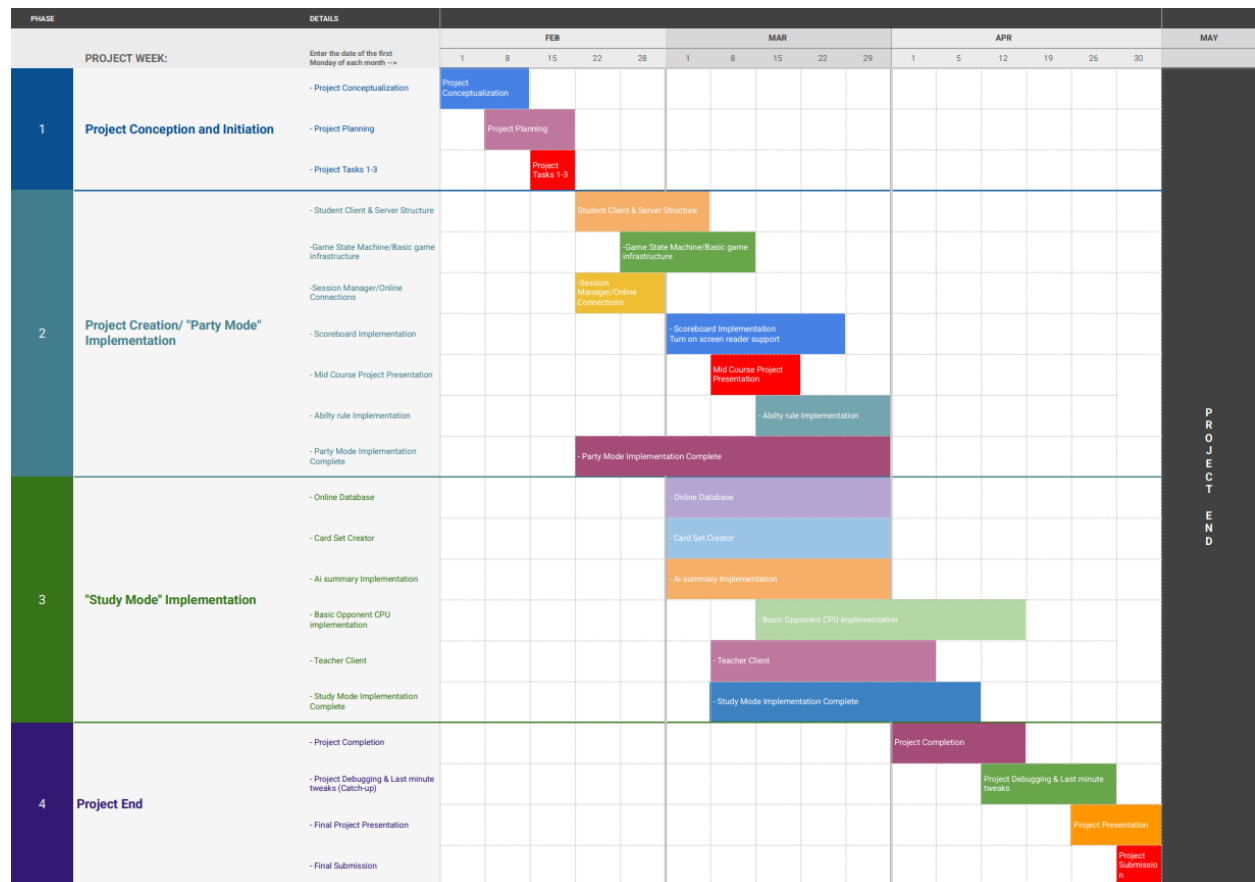


Problem Statement

In a classroom setting, it may be very hard for students to retain attention and individualized feedback that a teacher may not be able to give in a big classroom setting. Card Clash seeks to find a solution by allowing students to participate more through a game. Card Clash creates flash cards personalized to the user based on the topic at hand. It will achieve this by using AI to find patterns in their engagement and give feedback in a timely manner. The system will track gameplay and have users compete with their classmates. This information could also be shared with the instructor for better learning. Card Clash hopes to benefit classrooms seeking new learning opportunities with teachers that need data driven tools to help monitor performance.

Initial Timeline

Card Clash Timeline



System Requirements

Functional Requirements

Data Organization and Classification

FR-1 The system shall store questions in a database.

FR-2 The system shall classify questions by category or topic.

FR-3 The system shall store user profiles for students and teachers.

FR-4 The system shall track which questions have been assigned to each player within a session.

FR-5 The system shall prevent a player from receiving the same question more than once within a session.

FR-6 The system shall reset question assignment tracking after a session is finished.

Role-Based Functionality

FR-7 The system shall provide Teacher Mode with access to session creation, analytics, and reports.

FR-8 The system shall provide Student Mode with access to gameplay and individual metrics.

FR-9 The system shall restrict teacher-only functionality to authenticated teacher accounts.

Game Modes

FR-10 The system shall provide a Study Mode for singleplayer gameplay.

FR-11 The system shall provide a Party Mode for multiplayer gameplay.

FR-12 The system shall organize gameplay into sets of five rounds.

FR-13 The system shall allow a player to terminate a Study Mode session at any time.

FR-14 The system shall track time invested in Study Mode sessions.

Core Gameplay Mechanics

FR-15 The system shall assign each player a unique question within each round.

FR-16 The system shall randomly select questions from the question pool.

FR-17 The system shall begin each round with an Analyze Phase before answers can be submitted.

FR-18 The system shall allow players to request a card swap during the Analyze Phase.

FR-19 The system shall allow players to activate a card lock during the Analyze Phase.

FR-20 The system shall allow each player to use swap at most once per set.

FR-21 The system shall allow each player to use lock at most once per set.

FR-22 The system shall prevent a locked player's question from being swapped.

FR-23 The system shall resolve swap and lock actions simultaneously after the Analyze Phase ends.

FR-24 The system shall ensure swap outcomes do not result in duplicate questions within a round.

FR-25 The system shall start the Answer Phase timer only after swap and lock resolution is completed.

FR-26 The system shall allow one answer submission per player per round.

FR-27 The system shall validate answers against stored answer keys.

FR-28 The system shall assign points based on answer correctness.

FR-29 The system shall award double points for correctly answering a swapped question.

FR-30 The system shall update scores after each round.

User-Facing Features

FR-31 The system shall display player scores during gameplay.

FR-32 The system shall display a leaderboard for multiplayer sessions.

FR-33 The system shall display leaderboard rankings based on time spent in Study Mode.

FR-34 The system shall display leaderboard rankings based on subject performance.

Analytics and Reporting

FR-35 The system shall calculate accuracy metrics for each player.

FR-36 The system shall calculate performance metrics by category.

FR-37 The system shall identify high-performance and struggle areas for each player.

FR-38 The system shall store session analytics for later retrieval.

FR-39 The system shall generate AI-based summaries of class performance for teachers.

FR-40 The system shall display AI-generated summaries in the teacher interface.

Non-Functional Requirements

Performance

NFR-1 The system shall deliver real-time gameplay updates such as score, changes, timers, etc. to all connected clients within 1 second.

NFR-2 The system shall support up to 25 concurrent student users within a single multiplayer session.

NFR-3 The system shall complete swap and lock resolution within 1 second after the Analyze Phase ends.

NFR-4 The system shall generate AI-based session summaries within 15 seconds after completion.

Reliability and Data Integrity

NFR-5 The Game Server shall function as the authoritative source for all game state transitions.

NFR-6 The system shall ensure that question uniqueness constraints are maintained during concurrent player actions.

NFR-7 The system shall persist session data such that completed session results remain available after a system restart.

NFR-8 The system shall log gameplay events such as question assignment, swap / lock, scoring, etc., with timestamps to support auditing and debugging.

Security and Access Control

NFR-9 The system shall restrict Teacher Mode features to authenticated teacher accounts.

NFR-10 The system shall prevent students from accessing other student's performance metrics.

NFR-11 The system shall transmit authentication credentials and analytics data using secure communication protocols such as HTTPS.

Usability

NFR-12 A student shall be able to join a multiplayer session and begin gameplay within two minutes without previous experience.

NFR-13 The user interface shall clearly display the current gameplay phase and remaining time.

NFR-14 The system shall present leaderboard and performance data in a readable format suitable for classroom projection.

Scalability and Maintainability

NFR-15 The system architecture shall separate gameplay logic, data persistence, and analytics services to allow independent modification or scaling.

NFR-16 The system design shall support the addition of new question categories or game mechanics without requiring major redesign of the core architecture.

Availability and Network Tolerance

NFR-17 The system shall allow a temporarily disconnected user to reconnect to an active session and resynchronize with the current game state.

NFR-18 The system shall tolerate moderate classroom Wi-Fi variability without requiring session restart.

Architectural Design

Architectural Overview

Card Clash adheres to an authoritative client-server architecture designed to support real-time multiplayer gameplay, singleplayer study sessions, performance tracking, and AI-assisted reporting. The Game Server serves as the central authority. It manages game state, enforces gameplay rules, and ensures fairness during competitive interactions such as swap and lock mechanics.

The system is composed of five primary components:

1. Student Client
2. Teacher Client
3. Game server
4. Database
5. AI Summary Service

Real-time communication between clients and the server enables synchronized gameplay, while persistent database storage ensures that user profiles, question history, and analytics remain available beyond an individual session.

Major System Components

Student Client

The Student Client provides the user interface for both Study Mode (singleplayer) and Party Mode (multiplayer). Its responsibilities include:

- Displaying assigned questions
- Showing the current round phase
- Displaying timers and score updates
- Displaying leaderboards and personal performance metrics

The client does not determine game logic. All rule enforcement and scoring are handled by the Game Server to ensure fairness and consistency.

Teacher Client

The Teacher Client provides administrative and analytical functionality. Its responsibilities include:

- Creating and managing multiplayer sessions
- Viewing real-time class performance
- Viewing AI-generated summaries of class performance
- Monitoring topic-based strengths and struggle areas

Access to teacher functionality is restricted through authentication and role-based permissions.

Game Server (Authority)

The Game Server is the central coordinating component and maintains the authoritative game state. It is logically organized into the following modules:

Session Manager

- Creates and manages Study Mode and Party Mode sessions
- Organizes gameplay into sets of five rounds
- Tracks connected players and handles reconnections

Question Assignment Service

- Retrieves questions from the database
- Randomly selects questions from the pool
- Ensures no two players receive the same question within a round
- Prevents a player from receiving the same question more than once during a session
- Resets tracking after a session ends

Round Engine (State Machine)

Controls the lifecycle of each round:

1. Question assignment
2. Analyze Phase activation
3. Swap / lock submission window
4. Server-side swap / lock resolution
5. Start of Answer Phase timer
6. Answer validation
7. Scoring and leaderboard update

This structured state machine ensures synchronized transitions and consistent rule enforcement.

Ability Rules Engine

- Limits swap and lock usage to once per set
- Prevents swaps when a lock is active
- Ensures swap resolution does not violate uniqueness constraints
- Applies double-point scoring for swapped questions when answered correctly

Scoring and Leaderboard Engine

- Validates answers
- Calculates points
- Updates scores after each round
- Generates multiplayer leaderboards
- Computes Study Mode rankings based on time invested and subject performance

Metrics and Analytics Aggregator

- Calculates accuracy and performance by category
- Identifies high-performance and struggle areas
- Stores session analytics for future retrieval

Event Logger

- Records question assignments, swap / lock actions, answer submissions, and scoring outcomes
- Stores timestamped events for debugging and auditing
- Ensures consistency under concurrent player actions

Database

The Database stores:

- Question bank and category classifications
- Student and teacher profiles
- Session results and analytics
- Study Mode time invested
- Historical reports and leaderboard data

This persistent storage ensures completed session data remains available even after system restart.

AI Summary Service

The AI Summary Service receives aggregated session metrics from the Game Server and generates natural-language summaries for teachers.

These include:

- Overall class performance

- Category-level strengths and weaknesses
- Notable engagement and accuracy trends

The generated summary is returned to the server and displayed within the Teacher Client interface.

Communication and Data Flow

Real-Time Gameplay Communication

Student and Teacher Clients communicate with the Game Server using real-time messaging to:

- Synchronize round phases
- Submit swap and lock actions
- Submit answers
- Receive score and leaderboard updates

All gameplay decisions are processed server-side before being broadcast to clients.

Persistent Data Communication

The Game Server interacts with the Database to:

- Retrieve questions
- Store session results
- Maintain user profiles
- Store analytics and reports

AI Communication

After session completion, the Game Server sends aggregated performance data to the AI Summary Service and stores the generated summary for teacher access.

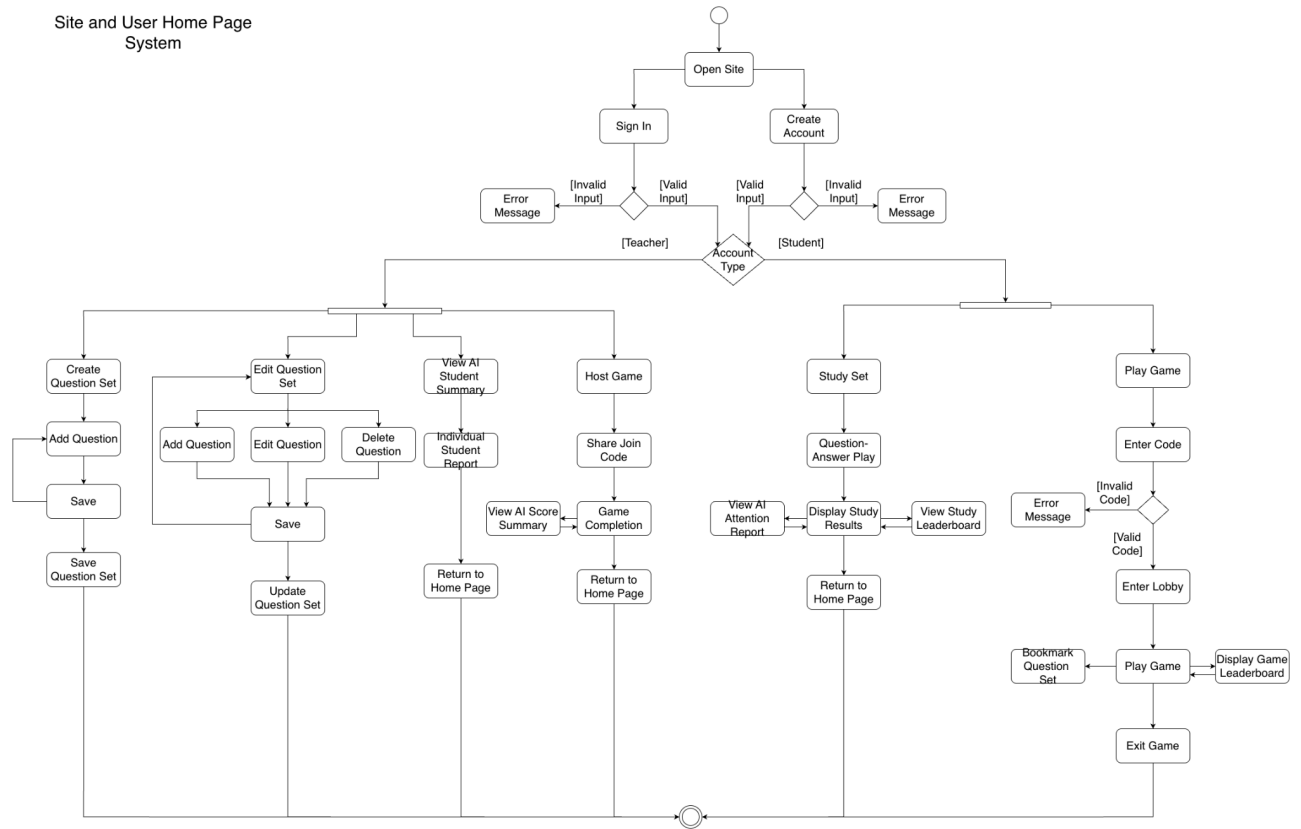
Architectural Traceability Summary

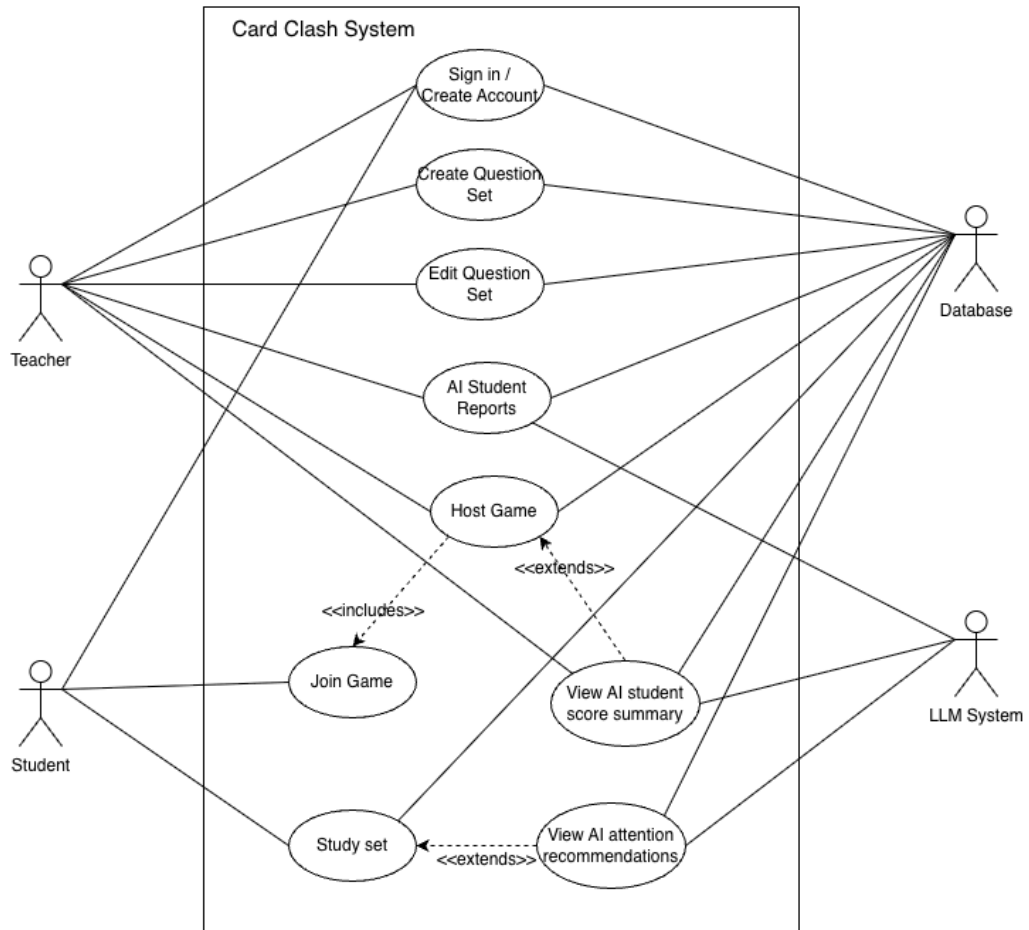
The architecture supports the system requirements in the following ways:

- Unique question assignment and session tracking are enforced by the Question Assignment Service.
- Swap and lock mechanics are handled within the Round Engine and Ability Rules Engine.
- Double-point scoring and leaderboard updates are handled by the Scoring Engine.
- Study Mode time tracking and category performance are handled by the Metrics Aggregator.
- Teacher-only functionality is protected through role-based access control.
- Real-time updates are supported through centralized server coordination.
- AI-generated summaries are supported through a dedicated analytics pipeline.

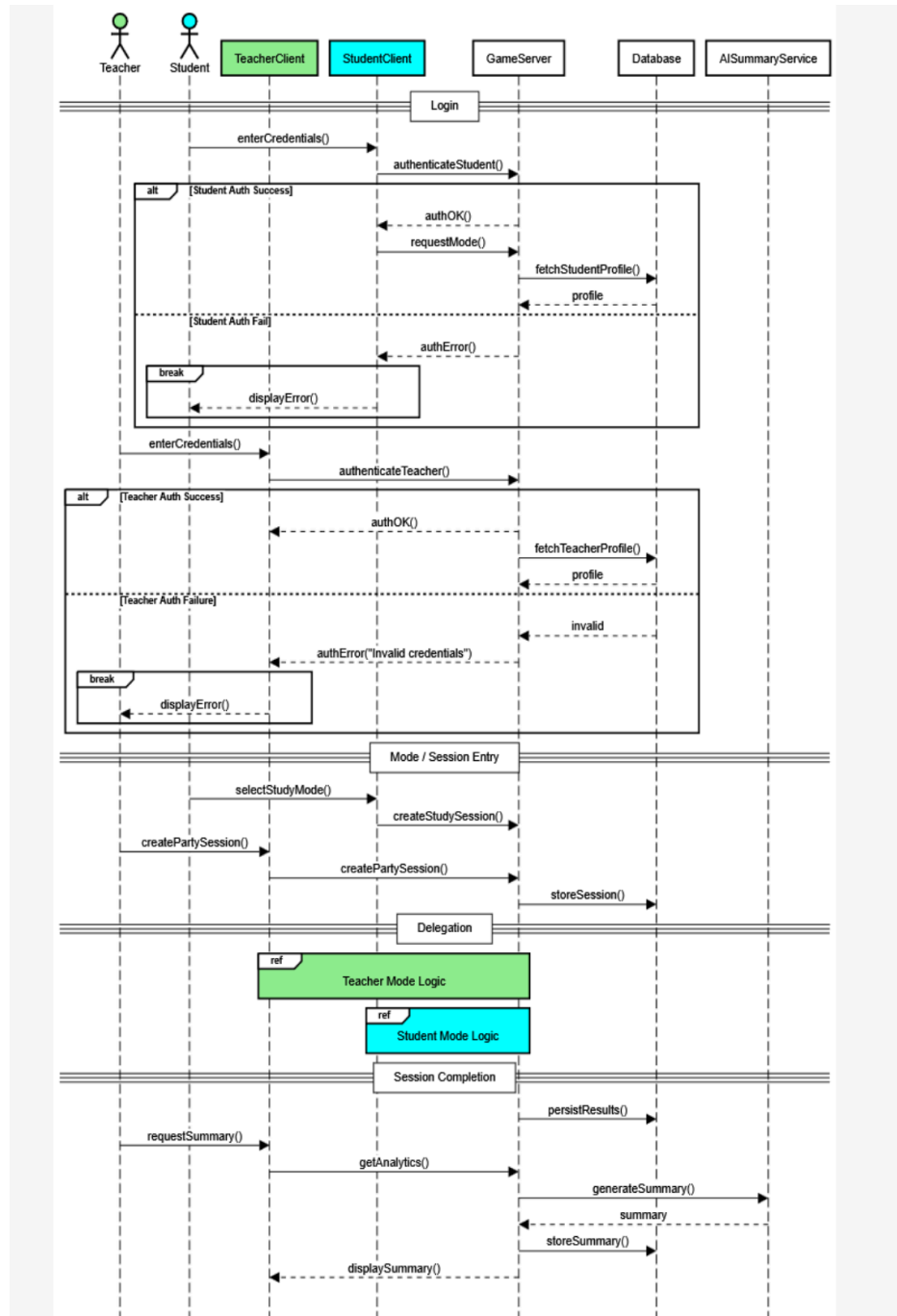
System Modeling

Site and User Home Page
System





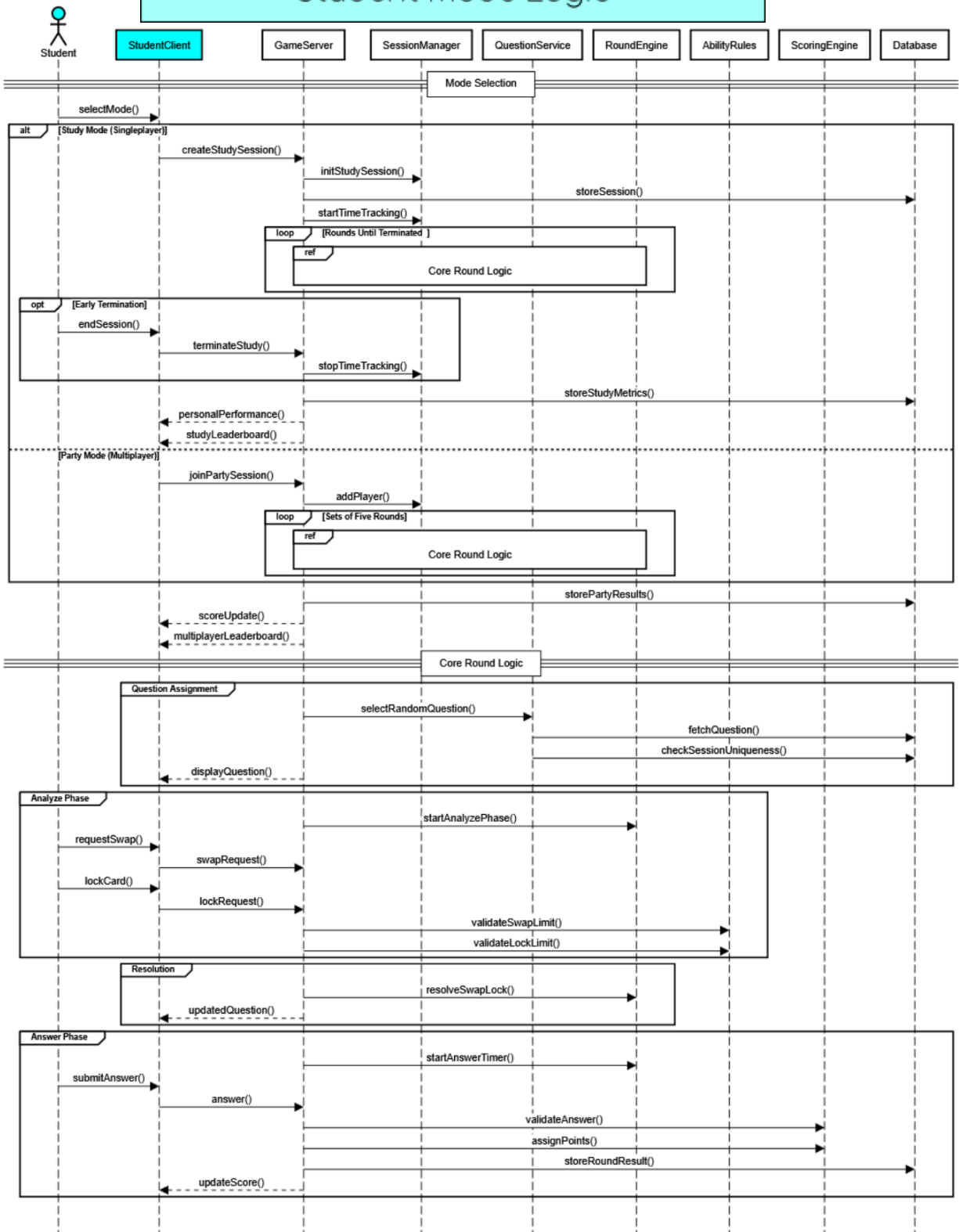
Sequence Diagram - Overview



***End of Sequence Diagram - Overview**

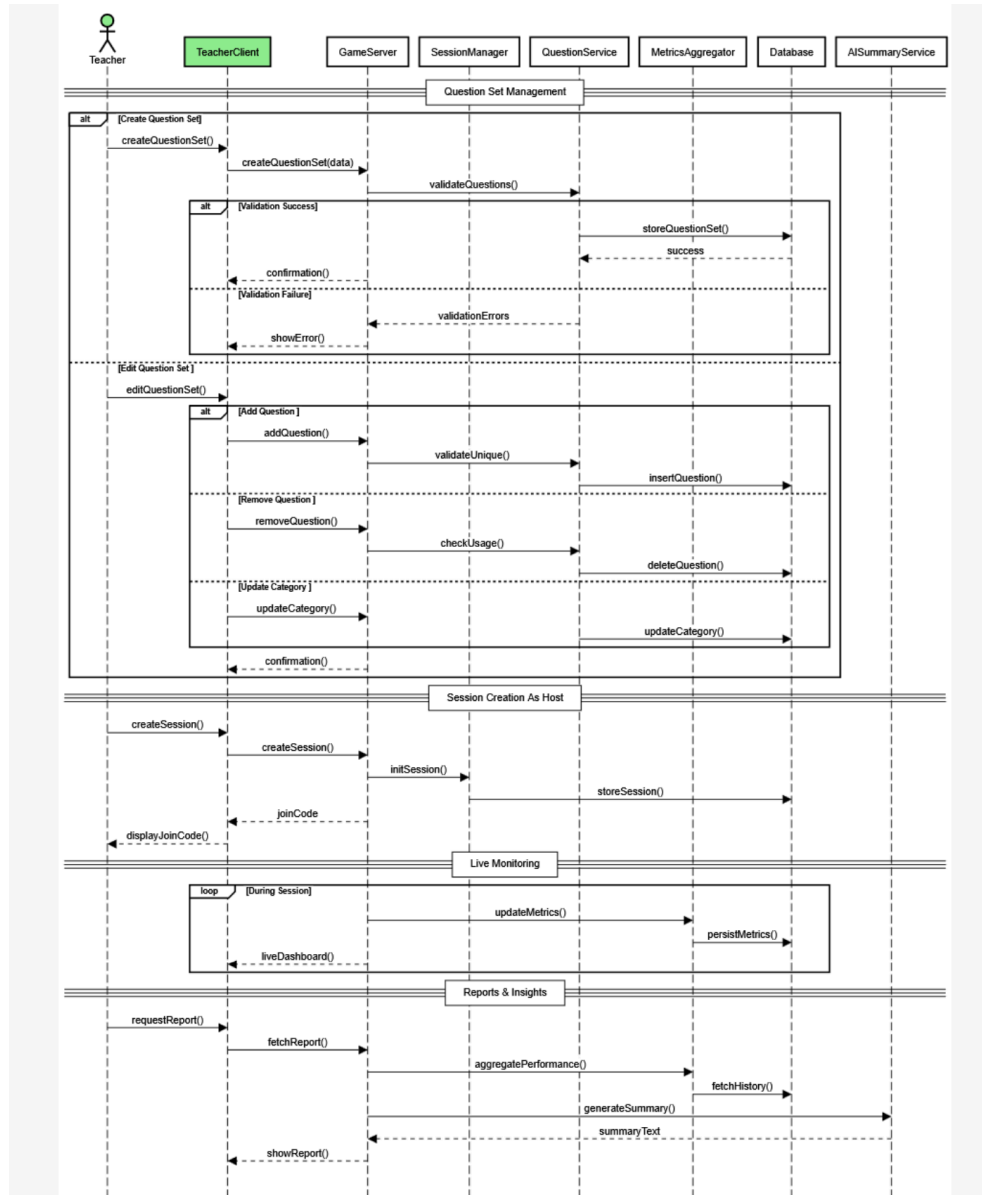
Sequence Diagram

Student Mode Logic



***End of Sequence Diagram - Student Mode**

Sequence Diagram Teacher Mode



***End of Sequence Diagram - Teacher Mode**

Class Diagram

